Exam Prep Lecture Scribe

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Generate a descriptive lecture scribe for exam preparation using the provided context. For every topic, reason through the definitions, step by step proofs, and examples to make the logical flow explicit for a student. Do not add outside material.

Here is a **descriptive lecture scribe for exam preparation** based strictly on the provided lecture slides. The logical flow, definitions, and reasoning are expanded step-by-step without adding external material.

Lecture 18 Scribe – Marginal PMF, Correlation, Covariance, Joint Gaussian RVs

(Exam Preparation Notes)

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1. Marginal PMF

1.1 Definition

The **key idea** behind marginal PMF is:

> The marginal PMF of one variable is obtained by summing the joint PMF over all possible values of the other variable.

Step-by-step reasoning

- We begin with a **joint PMF** $P(X, Y)$, which describes the probability of two variables together.
- However, often we are interested in only one variable (say X).
- To "remove" the effect of Y, we **sum over all possible values of** Y.

Thus:

- Marginal of X: sum over all y
- Marginal of Y: sum over all x

This process is called **marginalization**.

1.2 Marginal PMF from a Joint PMF Table

Logical procedure

1. Start with a **joint PMF table**.
2. To compute marginal PMF of X:
    - Fix a value of $X = x$
    - Sum all probabilities across corresponding row (over all Y)
3. To compute marginal PMF of Y:
    - Fix $Y = y$
    - Sum all probabilities down the column (over all X)

1.3 Example (Step-by-step reasoning)

Given a joint PMF:

Solution approach

1. Identify all possible values of X and Y
2. For each x, compute:
    - Sum of all joint probabilities where $X = x$
3. Repeat for each y

Conclusion

- The resulting values give:
    - $P_X(x)$: marginal PMF of X
    - $P_Y(y)$: marginal PMF of Y

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2. Correlation

2.1 Definition

A basic way to measure joint behavior is through the **product moment**:

- It gives the **average value of the product** XY

Interpretation

- This is called a **raw correlation measure**
- It captures how X and Y behave together

2.2 Limitations of Raw Correlation

The quantity:

$$R_{XY} = E[XY]$$

Step-by-step issues

1. It **depends on scale** of X and Y
2. It is computed **about the origin**, not about the mean
3. It mixes:
   - Mean values
   - Spread
   - Dependence

Important implication

- $R_{XY} = 0$ implies:
  - Variables are **orthogonal about the origin**
  - But not necessarily independent

2.3 Transition to Covariance

To fix the issue:

- We **center the variables** (subtract means)

This leads to a better measure: **Covariance**

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3. Covariance

3.1 Definition and Interpretation

Covariance is defined by:

- First subtract mean from each variable
- Then measure how they vary together

Step-by-step reasoning

1. Compute deviations:
   - $X - E[X]$
   - $Y - E[Y]$
2. Multiply deviations
3. Take expectation

3.2 Interpretation

Covariance tells us:

- Whether deviations from means:
  - Move **together** → positive covariance
  - Move **opposite** → negative covariance

3.3 Algebraic Form

Starting from definition:

- Expand the expression step-by-step
- Simplify using linearity of expectation

This leads to a simpler computable form.

3.4 Properties of Covariance

1. Symmetry

$$\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$$

2. Variance as Special Case

$$\mathrm{Var}(X) = \mathrm{Cov}(X, X)$$

3. Linearity w.r.t constants

- Scaling affects covariance proportionally

4. Independence

- If X and Y are independent:

$$\mathrm{Cov}(X, Y) = 0$$

⚠️ Important:

- Zero covariance ⇒ **uncorrelated**
- But does **NOT imply independence**

3.5 Example 1 (Step-by-step)

Solution procedure

1. Write covariance formula
2. Substitute given values
3. Expand expression
4. Use expectation properties
5. Simplify step-by-step

Final outcome

- Obtain numerical value of covariance after substitution and simplification

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3.6 Example 2 (Continuous Case)

Given a **joint PDF**, compute required quantities.

Step-by-step reasoning

1. Compute:
   - $E[X]$, $E[Y]$
2. Compute:
   - $E[XY]$
3. Substitute into covariance formula
4. Perform integrations carefully
5. Simplify result

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4. Pearson's Correlation Coefficient

4.1 Definition

- Measures the **effect of change in one variable when the other changes**

4.2 Interpretation

- Measures:
  - **Strength** of relationship
  - **Direction** of relationship

4.3 Key Properties

1. It is a **normalized version of covariance**
2. It is **unit-free**
3. It represents **linear relationship only**

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5. Jointly Gaussian Random Variables

5.1 Definition

A pair (X, Y) is **jointly Gaussian** if:

- The joint density has **bivariate Gaussian form**
- Marginals are also Gaussian

5.2 Intuition

- Many real-world variables result from:
  - Multiple small random effects
- Their combined behavior often becomes Gaussian

5.3 Real-Life Interpretation

Used in:

- Signal processing
- Communication systems
- Continuous noisy measurements

5.4 Example: Climate System

Variables:

- X: CO_2 concentration
- Y: Temperature anomaly

Reasoning

- Both influenced by many random processes:
  - Weather
  - Emissions
  - Ocean circulation
- Their joint behavior ≈ **Joint Gaussian distribution**

5.5 Algebraic Form and Interpretation

- The joint Gaussian distribution:
  - Captures **mean**
  - Captures **variance**
  - Captures **covariance (dependence)**

Final Summary (Exam Focus)

Key Takeaways:

- **Marginal PMF**: Sum out unwanted variable
- **Correlation $E[XY]$**: Raw measure, not centered
- **Covariance**: Measures joint variability around means
- **Zero covariance ≠ independence**
- **Pearson coefficient**: Normalized, unit-free correlation
- **Joint Gaussian**:
  - Fully described by mean + covariance
  - Widely used in modeling real systems

If you want, I can convert this into **LaTeX exam notes** or add **step-by-step solved numerical examples** for practice.

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Lecture 18 Scribe – Marginal PMF, Correlation, Covariance, Joint Gaussian RVs

(Exam Preparation Notes – Step-by-Step Reasoning)

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1. Marginal PMF

1.1 Definition

> The marginal PMF of one variable is obtained by summing the joint PMF over all possible values of the other variable.

Step-by-step reasoning

1. Start with a **joint PMF** $P(X, Y)$, which gives probabilities of pairs (x, y).
2. Suppose we are interested only in **one variable**, say X.
3. The probabilities in the joint PMF include the influence of Y.
4. To isolate X, we:
   - **Fix** $X = x$
   - **Sum over all possible values of** Y

Thus, we remove the dependency on Y by aggregation.

1.2 Marginal PMF from Joint PMF Table

Procedure

- Given a table of $P(X, Y)$:

To compute $P_X(x)$:

1. Select a fixed value x
2. Identify all entries where $X = x$
3. Sum all corresponding probabilities across different y

To compute $P_Y(y)$:

1. Fix y
2. Sum all entries across different x

1.3 Example (Logical Derivation)

Given a joint PMF:

Solution steps

1. List all values of X and Y
2. For each value of X:
   - Add probabilities across all Y
3. For each value of Y:
   - Add probabilities across all X

Conclusion

- The resulting sums form:
  - Marginal PMF of X
  - Marginal PMF of Y

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2. Correlation

2.1 Definition (Product Moment)

> A basic way to measure joint behavior is through the product moment.

- It computes:
  - The **average value of the product XY**

2.2 Interpretation

Step-by-step understanding

1. For each outcome:
   - Multiply X and Y
2. Take expectation:
   - This gives $E[XY]$
3. This reflects:
   - Whether large values of X occur with large values of Y

2.3 Limitations

The measure:

$$R_{XY} = E[XY]$$

Issues (step-by-step)

1. **Depends on scale**
   - If X or Y is scaled, value changes
2. **Computed about origin**
   - Does not consider mean values
3. **Mixes multiple effects**
   - Mean
   - Spread
   - Dependence

Special case

$$R_{XY} = 0 \Rightarrow \text{orthogonal about the origin}$$

2.4 Transition to Covariance

To fix these issues:

1. Remove the effect of mean
2. Center variables:
   - Replace X with $X - E[X]$
   - Replace Y with $Y - E[Y]$

This leads to **covariance**.

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3. Covariance

3.1 Definition

> We first subtract the mean from each variable and then measure how the centered quantities vary together.

3.2 Step-by-step Derivation

1. Compute mean values:
   - $E[X]$, $E[Y]$
2. Form deviations:
   - $X - E[X]$
   - $Y - E[Y]$
3. Multiply deviations:
   - $(X - E[X])(Y - E[Y])$
4. Take expectation:
   - This gives covariance

3.3 Interpretation

Covariance captures:

- **Positive covariance**:
  - Both variables deviate in same direction
- **Negative covariance**:
  - Variables deviate in opposite directions

3.4 Algebraic Form (Step-by-step)

Start with definition:

1. Expand product:

$$(X - E[X])(Y - E[Y])$$

2. Multiply terms
3. Apply expectation
4. Use linearity of expectation
5. Simplify

This gives a computationally simpler expression.

3.5 Properties of Covariance

1. Symmetry

$$\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$$

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2. Variance as Special Case

$$\mathrm{Var}(X) = \mathrm{Cov}(X, X)$$

---

3. Linearity w.r.t Constants

- Scaling variables scales covariance accordingly

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4. Independence

If X and Y are independent:

$$\mathrm{Cov}(X, Y) = 0$$

Important conclusion

- Zero covariance ⇒ **uncorrelated**
- But **does NOT imply independence**

3.6 Example 1 (Step-by-step)

Given

Expressions for X, Y

Solution steps

1. Write covariance formula
2. Substitute given expressions
3. Expand terms
4. Apply expectation operator
5. Simplify step-by-step

Conclusion

- Final value obtained after systematic substitution

3.7 Example 2 (Continuous Case)

Given a joint PDF.

Step-by-step solution

1. Compute:
   - $E[X]$ using integration
2. Compute:
   - $E[Y]$
3. Compute:
   - $E[XY]$
4. Substitute into covariance formula
5. Simplify result

---

4. Pearson's Correlation Coefficient

4.1 Definition

> Pearson's correlation coefficient calculates the effect of change in one variable when the other variable changes.

4.2 Step-by-step Interpretation

1. Start from covariance
2. Normalize using:
   - Standard deviation of X
   - Standard deviation of Y
3. This removes unit dependence

4.3 Properties

- Measures:
  - **Strength** of relationship
  - **Direction** of relationship
- It is:
  - **Unit-free**
  - **Normalized covariance**

---

5. Jointly Gaussian Random Variables

5.1 Definition

> A pair (X, Y) is jointly Gaussian if:

- The joint density has bivariate Gaussian form
- Marginals are Gaussian

5.2 Intuition

Step-by-step reasoning

1. Many real-world variables are influenced by:
   - Multiple small random factors
2. When combined:
   - They tend to follow Gaussian behavior
3. Hence, joint distribution becomes Gaussian

5.3 Applications

- Signal processing
- Communication systems
- Noisy measurements

5.4 Example: Climate System

Variables:

- X: CO_2 concentration
- Y: Temperature anomaly

Reasoning

1. Both depend on multiple random processes:
   - Weather
   - Ocean circulation
   - Emissions
2. These combined uncertainties:
   - Produce approximately Gaussian behavior
3. Hence:
   - Joint distribution ≈ Gaussian

5.5 Algebraic Form and Interpretation

- Joint Gaussian is characterized by:
  1. Means
  2. Variances
  3. Covariance

These parameters fully describe:

- Shape
- Spread
- Dependence structure

Final Exam Summary

Core Flow of Concepts

1. Joint PMF → Marginal PMF (by summation)
2. Raw correlation $E[XY]$ → not centered
3. Center variables → Covariance
4. Normalize covariance → Pearson coefficient
5. Extend to continuous case → Joint Gaussian RVs

Key Insights to Remember

- Marginalization removes dependence on one variable
- Covariance measures **joint deviation from mean**
- Zero covariance ≠ independence
- Pearson coefficient is **scale-free**
- Joint Gaussian models real-world continuous systems

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Lecture 18 Scribe – Marginal PMF, Correlation, Covariance, Joint Gaussian RVs

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1. Marginal PMF

1.1 Definition

> The marginal PMF of one variable is obtained by summing the joint PMF over all possible values of the other variable.

1.2 Step-by-step Understanding

1. We start with a **joint PMF** $P(X, Y)$, which gives probabilities of pairs (x, y).
2. This joint PMF captures the combined behavior of both variables.
3. Suppose we want information about only one variable, say X.
4. The probabilities in the joint PMF still depend on Y.
5. To remove this dependency:
   - Fix $X = x$
   - Sum over all values of Y

Thus, we obtain the **marginal PMF of** X.

1.3 Marginal PMF from Joint PMF Table

Procedure

To compute $P_X(x)$:

1. Fix a value x
2. Identify all entries where $X = x$
3. Add probabilities across all y

To compute $P_Y(y)$:

1. Fix y
2. Add probabilities across all x

1.4 Example (Step-by-step Reasoning)

Given a joint PMF:

Solution steps

1. Identify all values of X and Y
2. For each x:
   - Sum probabilities over all y
3. For each y:
   - Sum probabilities over all x

Result

- We obtain:
  - Marginal PMF of X
  - Marginal PMF of Y

---

2. Correlation

2.1 Definition (Product Moment)

> A basic way to measure joint behavior is through the product moment.

- It computes the **average value of the product XY**.

2.2 Step-by-step Interpretation

1. For every outcome:
   - Compute product XY
2. Take expectation:

- $E[XY]$

3. This gives a measure of how X and Y vary together

2.3 Limitations of $E[XY]$

$$R_{XY} = E[XY]$$

Step-by-step issues

1. **Depends on scale**
   - Changing units changes value
2. **Measured about origin**
   - Does not consider mean values
3. **Combines multiple effects**
   - Mean
   - Spread
   - Dependence

2.4 Special Case

$$R_{XY} = 0 \Rightarrow \text{orthogonal about the origin}$$

2.5 Transition

To remove the effect of mean:

1. Center variables:
   - $X - E[X]$
   - $Y - E[Y]$
2. This leads to **covariance**

---

3. Covariance

3.1 Definition

> Subtract the mean from each variable and measure how the centered quantities vary together.

3.2 Step-by-step Derivation

1. Compute means:
   - $E[X]$, $E[Y]$
2. Form deviations:
   - $X - E[X]$
   - $Y - E[Y]$
3. Multiply:
   - $(X - E[X])(Y - E[Y])$
4. Take expectation:
   - This gives covariance

3.3 Interpretation

- If both deviations:
  - Increase together → **positive covariance**
- If one increases and other decreases → **negative covariance**

3.4 Algebraic Form (Step-by-step)

1. Start with:

$$(X - E[X])(Y - E[Y])$$

2. Expand:
   - Multiply all terms
3. Apply expectation:
   - Use linearity
4. Simplify:
   - Obtain computational form

3.5 Properties

1. Symmetry

$$\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$$

---

2. Variance as Special Case

$$\mathrm{Var}(X) = \mathrm{Cov}(X, X)$$

---

3. Linearity

- Scaling affects covariance proportionally

---

4. Independence

If X and Y are independent:

$$\mathrm{Cov}(X, Y) = 0$$

Important Insight

- Zero covariance ⇒ **uncorrelated**
- Does **not imply independence**

3.6 Example 1 (Step-by-step)

Given

Expressions for variables

Solution steps

1. Write covariance formula
2. Substitute given expressions
3. Expand algebraically
4. Apply expectation operator
5. Simplify step-by-step

Final Step

- Obtain final covariance value after simplification

3.7 Example 2 (Continuous Case)

Given joint PDF of (X, Y)

Step-by-step solution

1. Compute:

    - $E[X]$ using integration
2. Compute:
    - $E[Y]$
3. Compute:
    - $E[XY]$
4. Substitute into covariance formula
5. Simplify integrals

---

4. Pearson's Correlation Coefficient

4.1 Definition

> Pearson's correlation coefficient calculates the effect of change in one variable when the other variable changes.

4.2 Step-by-step Understanding

1. Start with covariance
2. Normalize using:
    - Standard deviation of X
    - Standard deviation of Y
3. This removes unit dependence

4.3 Properties

- Measures:
    - Strength of relationship
    - Direction of relationship
- It is:
    - **Unit-free**
    - **Normalized covariance**

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5. Jointly Gaussian Random Variables

5.1 Definition

> A pair (X, Y) is jointly Gaussian if:

- Joint density is bivariate Gaussian
- Marginals are Gaussian

5.2 Step-by-step Intuition

1. Many real-world variables are affected by:
   - Multiple small random factors
2. These combine to produce:
   - Gaussian-like behavior
3. Hence, joint distribution becomes Gaussian

5.3 Applications

- Signal processing
- Communication systems
- Continuous noisy measurements

5.4 Example: Climate System

Variables:

- X: CO_2 concentration
- Y: Temperature anomaly

Reasoning

1. Influenced by multiple random processes:
   - Solar radiation
   - Ocean circulation
   - Emissions
2. Combined randomness:
   - Leads to Gaussian behavior
3. Hence:
   - Joint Gaussian model is appropriate

5.5 Algebraic Form and Interpretation

- Fully described by:
  1. Mean values
  2. Variances
  3. Covariance

These parameters determine:

- Shape
- Spread
- Dependence structure

Final Exam Summary

Concept Flow

1. Joint PMF → Marginal PMF (by summation)
2. Raw correlation $E[XY]$ → not centered
3. Center variables → Covariance
4. Normalize → Pearson coefficient
5. Extend to continuous → Joint Gaussian

Key Takeaways

- Marginal PMF removes dependence on other variables
- Covariance measures joint variation around mean
- Zero covariance ≠ independence
- Pearson coefficient is scale-independent
- Joint Gaussian models real-world continuous systems

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